

6.15 & 6.16 Complete Student Project & Ethical Use

طلبہ گروپوں میں ایک چھوٹا، مقامی اور آسان منصوبہ مکمل کرتے ہیں جس میں مسئلے سے پروٹوٹائپ، رائے، ایم وی پی
منصوبہ...

CS XII

Chapter 6: Entrepreneurship in the Digital Age

Lecture 16 of 16

6.15 & 6.16 Complete Student Project & Ethical Use

LEARN IT

You will work in **groups** and complete one small entrepreneurship project. The project should be **simple, local, and possible in the classroom**.

The purpose is to practise the **complete journey**: problem → prototype → feedback → MVP plan → risky assumption → beachhead market.

Suggested project ideas: homework and test reminder system, college notice organizer, canteen pre-order system, small shop WhatsApp catalogue, college van update system, lost and found item reporting system, study notes organizer.

Table 6.12: Student project sheet (template)

Item	Your group's answer
Problem statement	
Target users	
Business idea	
Low-fidelity prototype description	
Feedback from users	
Improvement made after feedback	
MVP plan	
Riskiest assumption	
How to test the riskiest assumption	
Beachhead market	
Marketing method for first users	

WORKED EXAMPLE

Worked example: a completed project sheet (Lost and Found system).

Item	Answer
Problem statement	Students of our college in Mirpurkhas lose items (bottles, calculators, ID cards) and rarely get them back because there is no central place to report them.
Target users	Students and staff of our college.
Business idea	A Lost and Found WhatsApp group plus a Google Form to report lost or found items.
Low-fidelity prototype	Two paper screens: "Report lost item" (item, colour, place, phone of class rep) and "Found items list" with sketches.
Feedback from users	6 of 8 testers asked for photos; 3 did not want their own phone number shown.
Improvement made	Added a photo upload; replaced personal numbers with "collect from the library desk".
MVP plan	Google Form + one WhatsApp group run by two volunteers; free; tested for two weeks.
Riskiest assumption	Students who find items will take the trouble to report them.
How to test it	Count how many "found" reports arrive in two weeks compared with "lost" reports.
Beachhead market	Grade XII students of our college.
Marketing method	Announcement in morning assembly, a poster at the library and a message in each class group.

Table 6.13: Rubric for project assessment

A **rubric** is a marking guide that describes what excellent, satisfactory and weak work looks like for each part of a project.

Criteria	Excellent	Satisfactory	Needs Improvement
Problem ID	Clear, local, and important.	Understandable but needs detail.	Vague or too broad.
Prototype	Shows clear flow and features.	Shows the basic idea.	Incomplete or unclear.
Feedback & Fixes	Collected properly; changes clearly based on feedback.	Collected but analysis is weak; some changes made.	No clear feedback collected; changes not explained.
MVP Plan	Essential features and clear testing plan.	Basic features but needs testing clarity.	Confused with the final product.

REMEMBER IT

Use the rubric **before** you submit: read your project sheet row by row and ask, "Is this Excellent yet?" For "Feedback & Fixes", always write **which** feedback caused **which** change.

COMMON MISTAKE

Mistake: Writing an MVP plan that lists every feature of the dream app. The rubric marks this as "Confused with the final product".

Fix: Keep the MVP to the essential feature(s) and add a clear testing plan (how many users, how long, what you will measure).

اردو میں سمجھیں

طلبہ گروپوں میں ایک چھوٹا، مقامی اور آسان منصوبہ مکمل کرتے ہیں جس میں مسئلے سے پروٹوٹائپ، رائے، ایم وی پی منصوبہ، سب سے خطرناک مفروضہ اور بیچ بیڈ مارکیٹ تک پورا سفر شامل ہوتا ہے۔ منصوبے کو جانچنے کے لیے روبرک (Rubric) استعمال ہوتا ہے جو مسئلے کی پہچان، پروٹوٹائپ، رائے و بہتری اور ایم وی پی منصوبے کو بہترین، تسلی بخش یا بہتری کی ضرورت کے درجوں میں پرکھتا ہے۔

CHECK YOURSELF

1. Name four suggested project ideas.
2. What does "Excellent" look like for Problem ID in the rubric?
3. Which rubric level says "Confused with the final product"?

Answers: 1. Any four: homework and test reminder system, college notice organizer, canteen pre-order system, small shop WhatsApp catalogue, college van update system, lost and found reporting system, study notes organizer. 2. Clear, local, and important. 3. Needs Improvement (for the MVP Plan).

LEARN IT

Digital entrepreneurship should be **responsible**. When collecting user feedback, you must respect:

- **Privacy:** people's personal information belongs to them.
- **Truthfulness:** report results honestly; do not make up numbers or hide bad feedback.
- **Fairness:** treat all users equally.

You should **not collect unnecessary personal information** or **share phone numbers or photos without permission**.

Equity point

A good digital solution should consider users who have **weak internet**, **limited devices**, or **language difficulty**. A design with **large buttons**, **simple Urdu/Sindhi instructions**, and **low data usage** helps ensure **equal access**. *Equity* means making sure everyone, including people with fewer resources, can use the solution.

WORKED EXAMPLE

Worked example: fix an unethical feedback form. A team's canteen feedback form asks:

1. Full name, father's name, CNIC/B-form number, home address. → **Problem:** unnecessary personal data. **Fix:** ask only class (or nothing personal at all).
2. "Upload your selfie." → **Problem:** photos are not needed and could be misused. **Fix:** remove.
3. The team plans to post all respondents' phone numbers in the class group. → **Problem:** sharing numbers without permission. **Fix:** never share; keep responses private to the team.
4. The form is in English only and has a 5 MB video at the top. → **Problem:** excludes users with language difficulty or weak internet. **Fix:** add Urdu/Sindhi instructions and remove the heavy video.

When presenting, the team reports "30 out of 40 would use it again" exactly, without rounding it up to "almost everyone". That is truthfulness.

REMEMBER IT

Ask yourself "P-T-F-E": Is it **P**rivate, **T**ruthful, **F**air, and **E**qual-access?

COMMON MISTAKE

Mistake: "More data is always better, so let's collect everything."

Fix: Collect only what you need to answer your test questions. Extra personal data creates risk for users and gives you no benefit.

اردو میں سمجھیں

ڈیجیٹل کاروبار ذمہ داری سے کرنا چاہیے۔ رائے جمع کرتے وقت لوگوں کی رازداری (Privacy)، سچائی اور انصاف کا خیال رکھیں؛ غیر ضروری ذاتی معلومات نہ لیں اور اجازت کے بغیر فون نمبر یا تصاویر شیئر نہ کریں۔ اچھا حل کمزور انٹرنیٹ، محدود موبائل اور زبان کی مشکل والے لوگوں کا بھی خیال رکھتا ہے، جیسے بڑے بٹن، آسان اردو یا سندھی ہدایات اور کم ڈیٹا کا استعمال۔

CHECK YOURSELF

1. Name the three values students must respect when collecting feedback.
2. Give two things students must not do with users' information.
3. List three design features that support equal access.

Answers: 1. Privacy, truthfulness, fairness. 2. Collect unnecessary personal information; share phone numbers or photos without permission. 3. Large buttons, simple Urdu/Sindhi instructions, low data usage.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. The student project should be:

- a. Small, local, possible in class time
- b. A full bank app
- c. A new OS
- d. A satellite

MCQ 2. It should include:

- a. Problem, idea, prototype/MVP, test, ethics
- b. Only a logo
- c. Only careers
- d. Only a Python hello

MCQ 3. Ethical use of digital tools means:

- a. Consent, no harassment, no leaked data, honest claims
- b. Scrape classmates' photos
- c. Fake testimonials
- d. Spam the college group

MCQ 4. The rubric rewards:

- a. A real test, not a fantasy pitch deck
- b. Unicorn valuation
- c. English poetry only
- d. NAND counts

MCQ 5. Safe tools:

- a. Forms/docs you have a right to use; no piracy
- b. Cracked Adobe
- c. Stolen question papers
- d. Phishing kits

MCQ 6. 6.15 and 6.16 together remind you:

- a. Build something small and don't harm people
- b. Skip users
- c. Skip tests
- d. Skip the write-up

Extra short questions

1. List the project pieces the book wants.
2. Two ethical rules for digital tools.
3. What "possible" means.
4. A fail on ethics even if the idea is clever.

Extra long question

1. Write a one-page project proposal that would score well: 8 headings including ethics and a 7-day test. Invent a local idea that is not the canteen copy-paste.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Team, local problem, idea, prototype, MVP or test, beachhead, results, ethics, reflection.
2. Permission for photos; no copyright theft; no unsafe personal data in public sheets.
3. You can test this week with people you can walk to.
4. Secretly recording students in the canteen.

Long

1. e.g. lab-equipment booking. Headings match rubric. Ethics: no shame list of late students on a public wall. Metric: 10 bookings honoured.

Next lecture: Chapter exam (in the chapter lecture PDF)